Nonlinear Transmission of Monetary Policy and Housing Market Imbalances: Evidence from a Factor-Augmented Threshold VAR Analysis

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Abstract

In recent decades, persistently low rates have driven housing booms and prompted questions on how market imbalances interact with monetary policy. In this paper, I investigate how such imbalances affect the monetary transmission in the United States. I create a boom-bust cycle indicator from the rent-price ratio and credit-to-GDP gap, then embed it in a factor-augmented threshold VAR with two regimes to isolate the two parts of the cycle. Impulse responses show that monetary policy itself behaves differently conditional on the state of the housing market. Interest rates are eased much more gradually around booms, which suggests that broader macro-financial signals are incorporated into the policy framework in order to effectively manage deleveraging in the market. The transmission channel appears to be less affected, as real and financial aggregates show only somewhat larger contractions in response to a monetary policy shock in a booming market.

Keywords: Housing market, Monetary policy, Threshold Vector Auto-Regression, Latent factors, Generalized Impulse Responses

JEL: E32, E44, E52, C32

Introduction

Housing is a central driver of business cycles. Residential investment makes up a sizable share of demand, and the housing market amplifies shocks through credit and monetary channels. After the Global Financial Crisis, prolonged low rates produced a large expansion in mortgage lending and a sharp rise in property prices, raising bubble concerns. As inflation rose sharply in 2022 and remained persistently high, central banks entered prolonged tightening cycles. Although many have since concluded these cycles and begun easing monetary conditions, reassessing the relationship between the housing market and monetary policy remains a pressing issue for researchers and policymakers.

The motivation for this study stems from both empirical and policy-related challenges. First, a deeper understanding of the interaction between housing markets and monetary policy is essential, as housing is a leading indicator of broader economic imbalances. Historical episodes, such as the housing price bubble identified by Case and Shiller (2003) and subsequent market corrections illustrate how sustained deviations from fundamental values can lead to severe economic downturns. Kydland et al. (2016) provides empirical evidence that residential investment is not only a leading indicator, but also a critical driver of business cycle fluctuations. Adelino et al. (2018) surveys the role that mortgages and inflated house prices played in the GFC, and the literature review by Duca et al. (2021) compiles evidence that housing cycles are closely linked to broader business cycle dynamics at the international level.

Second, there remains considerable ambiguity regarding the nonlinear transmission mechanisms through which monetary policy shocks affect the economy. Li and St-Amant (2010) investigate Canadian monetary transmission and find strong state-dependent dynamics in the presence of financial stress. Fry-Mckibbin and Zheng (2016) report similar findings for U.S. monetary transmission during crises. Schmidt (2020) examines nonlinear monetary transmission in European asset markets and also finds stronger responses during periods of high stress. Traditional linear models may overlook important regime-dependent dynamics, particularly how shocks interact with underlying market vulnerabilities.

Third, the study is motivated by the three-way interaction between housing markets, monetary policy, and financial conditions. The seminal work of Bernanke et al. (1999) on the financial accelerator illustrates how credit market imperfections can magnify the effects of monetary shocks. Papers by Iacoviello (2005), Iacoviello and Minetti (2008), and Mian and Sufi (2011) examine the interaction between housing, monetary policy, and borrowing constraints. These studies show that house price fluctuations can amplify aggregate demand via collateral effects that influence borrowing capacity.

Although the low interest rate environment post-2008 encouraged lending and inflated asset prices, the subsequent rise in inflation forced a reversal in monetary policy. This tightening may have implications for speculative pressures in housing markets. When central banks raise rates in a highly leveraged and credit-sensitive economy, prior research suggests that the impact may be amplified under high-stress conditions. By introducing a state-dependent Threshold Vector Autoregressive (TVAR) framework, this study aims to identify and quantify such nonlinearities. In doing so, it contributes to the literature on the heterogeneous effects of policy interventions in overheated versus stable market conditions, as well as on the evolution of housing market imbalances. Rising home prices, in turn, have been shown to be closely linked to financial stress and to serve as early warning indicators of broader economic downturns.

This paper presents a natural continuation of the existing literature. Using the rent-price ratio and the credit-to-GDP gap as measures of real estate and credit market imbalances, I construct a synthetic index of to track the boom-bust cycle in the housing market using principal component analysis (PCA). This composite indicator is embedded into a Factor-Augmented TVAR (TFAVAR) model to identify forming bubbles in the real estate market and to trace regime-specific impulse responses to monetary shocks across a carefully selected set of variables. These impulse responses are estimated using Generalized Impulse Responses that account for regime switches. The results indicate some regime-dependent effects of monetary policy shocks, although the differences across regimes are relatively minor. Importantly, the behavior of monetary policy appears to shift significantly depending on the state of the economy. During housing booms, the policy stance remains tighter for longer, and interest rates adjust more gradually, reflecting the Federal Reserve's cautious approach to navigating financial and housing market imbalances. These findings suggest that monetary policy responds to broader economic conditions than those captured by the standard Taylor rule, particularly when managing deleveraging during housing booms.

The remainder of this paper is structured as follows. Section 2 describes the dataset, which comprises quarterly U.S. data from 1970Q2 to 2024Q3 and includes real aggregates, financial stress measures, and housing market variables. Section 3 outlines the methodological framework, including the construction of the composite housing stress indicator and the estimation of the TVAR model. Section 4 presents the empirical results, focusing on the nonlinear transmission of monetary policy shocks. Section 5 concludes with a discussion of the implications and directions for future research.

Data

To fully capture complex macro-financial dynamics, I construct a quarterly dataset consisting of real aggregates, financial stress indicators, and housing market variables for the United States from 1970Q2 to 2024Q3. The dataset contains 10 variables: Real GDP (Y), Consumer Prices (P), Real Property Prices (HP), Rent-price ratio (RP), Real Estate Loans (HL), Mortgage Rate (MR), Federal Funds Rate (R), Term Spread (TS), Credit Spread (CS), and Credit-to-GDP Gap (CGG).

To account for the continuity of monetary policy action during periods when nominal rates are constrained by the Zero-Lower-Bound, the Wu and Xia (2016) shadow rate is used in place of the observed Federal Funds Rate values. The term spread, calculated as the yield on 10-year minus the yield on 1-year Treasury bills, serves as a measure of expectations about future interest rate fluctuations. The credit spread is calculated as the return on BAA minus AAA-rated corporate bonds and is useful for tracking episodes of financial stress or tight financial conditions.

The rent-price ratio serves as a valuation metric of the housing market aimed at capturing potential misalignments between property prices and economic fundamentals.

I calculate the rent-price ratio as $\ln(\text{Real Rental Index}) - \ln(\text{Real Property Prices Index})$, where the Real Rental Index is $\frac{\text{CPI: Rent of primary residence}}{\text{CPI: All items less shelter}}$. The Credit-to-GDP gap serves as an indicator of credit cycle imbalances and helps detect periods of excessive lending relative to long-run trends. CGG is defined as the deviation of $\frac{\text{Credit}_t}{\text{GDP}_t}$ from its long-run trend and is calculated using an HP filter.

All data except the CGG series are downloaded from the FRED database. The CGG series is downloaded from the BIS data portal. Variables Y, P, HP, and HL are log-normalized. All variables except R are scaled to zero mean and unit standard deviation. Table 1 below summarizes the variables and transformation steps:

Variable	Symbol	Definition / Calculation	Transformation	Data Source
Real GDP	Y	Real Gross Domestic Product	Log-level, scaled to zero mean and unit std.	FRED
Consumer Prices	P	Consumer Price Index	Log-level, scaled to zero mean and unit std.	FRED
Real Property Prices	HP	Real Property Prices Index	Log-level, scaled to zero mean and unit std.	FRED
Rent-Price Ratio	RP	$\ln(HP) - \ln\left(\frac{\text{CPI: Rent of primary residence}}{\text{CPI: All items less shelter}}\right)$	Scaled to zero mean and unit std. dev.	FRED
Real Estate Loans	HL	Total Real Estate Loans	Log-level, scaled to zero mean and unit std.	FRED
Mortgage Rate	MR	Mortgage interest rate	Levels, scaled to zero mean and unit std.	FRED
Federal Funds / Shadow Rate	R	Nominal Federal Funds Rate; replaced with shadow rate during ZLB periods	Entered in levels	FRED
Term Spread	TS	Difference between 10-year and 1-year Treasury yields	Levels, scaled to zero mean and unit std.	FRED
Credit Spread	CS	Difference between BAA and AAA corporate bond yields	Levels, scaled to zero mean and unit std.	FRED
Credit-to-GDP Gap	CGG	Deviation of $\frac{\text{Credit}_t}{\text{GDP}_t}$ from trend (HP filter)	Scaled to zero mean and unit std. dev.	BIS

Table 1: Variable definitions and transformations

Methodology

This section outlines the analytical framework used in the paper. The methodological challenges are threefold: (i) credibly identifying periods of booms and busts in the housing market; (ii) modeling nonlinear dynamics; and (iii) correctly capturing the propagation of monetary policy shocks in a nonlinear system. Threshold-type models are well-established for handling nonlinearities in empirical macroeconomics (see B. Hansen (1999), B. E. Hansen (2000), Lo and Zivot (2001), Caner and Hansen (2001), B. E. Hansen and Seo (2002), B. E. Hansen (2011)). Generalized impulse response functions (GIRFs), introduced by Koop et al. (1996), offer a way to analyze dynamic responses in such settings. I closely follow the GIRF implementation methods of Andreasen et al. (2021). Lastly, composite indicators are widely used in empirical macro for tracking financial stress or business cycle conditions (see Estrella and Mishkin (1998), Stock and Watson (1989), Bai and Ng (2002), Union and Centre (2008), Hatzius et al. (2010), Koop and Korobilis (2014)).

Identifying the boom-bust cycle in the Housing Market

Speculative pressures in housing markets tend to arise from the interaction of credit expansion and housing overvaluation. I propose a framework that captures this joint dynamic using two core measures: the credit-to-GDP gap (CGG), which signals excessive credit growth, and the rent–price ratio (RP), which reflects deviations of housing prices from rental fundamentals. While each measure is informative on its own, they are susceptible to measurement error and idiosyncratic variation. To synthesize their shared signal and reduce noise, I construct a composite indicator using principal component analysis (PCA).

The use of such composite indicators is well grounded in the literature. For example, Borio and Drehmann (2009) and Drehmann and Juselius (2014) aggregate various imbalance indicators into early warning measures for financial crises. Similarly, Alessandri and Mumtaz (2019) employ a nonlinear VAR framework centered around a composite financial stress index, and Auerbach and Gorodnichenko (2012) use a smooth transition VAR based on a composite business cycle indicator.

Construction of the housing boom-bust cycle indicator begins with standardized series (as seen in the Data section): both the CGG and RP series are scaled to have zero mean and unit variance, since PCA is scale-sensitive. The standard PCA procedure involves decomposing the covariance matrix of these inputs. The eigenvector corresponding to the largest eigenvalue provides the linear combination that maximizes shared variance:

$$PC1 = w_1 CGG + w_2 RP \tag{1}$$

where w_1 and w_2 are the weights from the leading eigenvector. This first principal component (PC1) thus captures the joint variation in credit and housing price signals, functioning as a latent factor that represents the simultaneous emergence of credit excess and housing overvaluation. The resulting time series is shown in Figure 1.

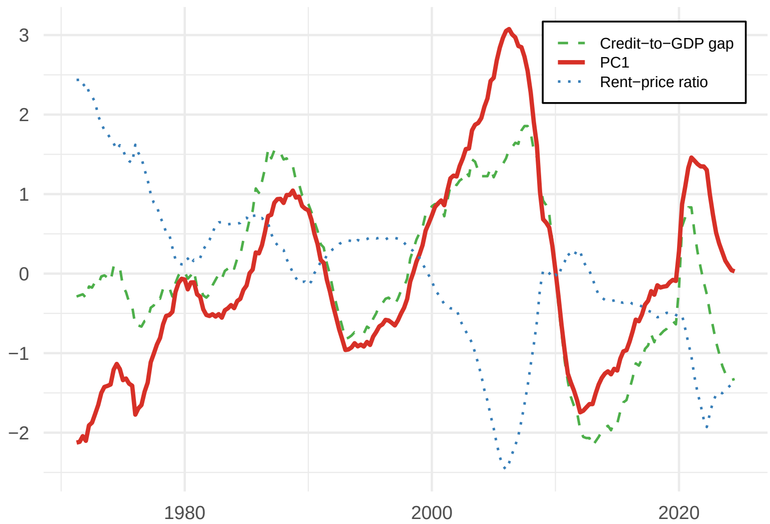

Figure 1: An indicator of the boom-bust cycle in the housing market.

PC1 explains roughly 61% of the shared variance. This leaves 39% of the variation unexplained, which may raise concerns about missing relevant dynamics. However, a visual inspection suggests that the index rises precisely when credit expansion accelerates and house prices decouple from fundamentals. It therefore provides a theoretically grounded and practically useful indicator of boom-bust cycles in the housing market.

Threshold VAR Estimation and Identification

Nonlinear models are becoming staples in empirical macroeconomics, especially when assessing how aggregate dynamics shift during episodes of financial strain, heightened uncertainty, or outright recessions. For example, Schüler (2014) and Lhuissier et al. (2016) show that uncertainty shocks matter far more when the banking sector is already under pressure. Likewise, Chiu and Hacioglu Hoke (2016) finds that financial shocks in downturns spark disproportionately larger contractions, while Ferraresi et al. (2015) and Afonso et al. (2018) use TVAR frameworks to uncover that fiscal multipliers are larger when credit conditions tighten. More recently, Kole and Dijk (2023) finds that financial stress shocks have stronger impacts in bearish markets and during recessionary times, and Mittnik and Semmler (2018) document how leverage cycles can destabilize macro dynamics.

Across these studies, scholars have adopted a variety of empirical models to capture nonlinear behavior. The threshold VAR (TVAR) stands out for its transparency and modest computational demands. Unlike Markov-switching VAR (MSVAR) and Time-Varying Parameter VAR (TVP-VAR) models, which capture regime changes via latent processes or gradually evolving coefficients, the TVAR explicitly splits the sample based on an observed variable. The explicit regime separation makes the interpretations very clear, as the differing propagation of shocks can be directly attributed to the state of the economy as identified by the regimes. The straight forward interpretation as well as the relative computational ease makes the TVAR and ideal choice for modeling the nonlinear transmission of monetary policy across the housing cycle.

Naturally, no empirical method is without its drawbacks. Firstly, the state of the economy is more accurately represented by a smooth spectrum, rather than black and white worldview of the binary regime indicator suggested by the TVAR. Secondly, the TVAR substantially reduces degrees of freedom, as it essentially doubles the number of estimated parameters (assuming there are two regimes).

As shown in the Results Section, the regime switches are not at all abrupt, and the regimes do indeed have a clear interpretation, as such I see no reason to address the first shortcoming of the TVAR model by introducing smoother transitions. As for the issue of over-parametrization, I adopt a dimensionality-reduction strategy using Factor-Augmented VARs (FAVAR), originally proposed in Bernanke et al. (2005).

A standard FAVAR can be estimated using two steps. First, a small number of latent factors are extracted (typically) from a large panel of observed variables. These factors can be extracted using different techniques, such as principal components or dynamic factor models. For the purposes of this paper I rely on PCA to extract the factors. Second, a VAR model is fit on variables (typically) excluded from the factor extraction step and the extracted latent factors. This two stage approach has the benefit significantly reducing dimensionalty, while retaining the majority of the shared variance across a number of variables. As the second step suggests, the FAVAR at its core is a standard VAR estimated using latent factors, which makes its combination with the TVAR estimation steps straight forward.

Although FAVARs often draw on hundreds of series, my threshold FAVAR (TFAVAR) relies on just seven indicators (Y, P, HP, HL, MR, TS, and CS) in the factor extraction process. While unusual, this streamlined setup remains consistent with the original motivation behind the FAVAR model: overcoming the standard VAR's curse of dimensionality without sacrificing informational richness. By focusing on these seven series, I preserve a relatively parsimonious framework that captures the main transmission channels of monetary policy while also incorporating aggregates relevant for the housing market.

I extract three principal components, which jointly explain around 93% of the variance across the seven variables, as shown in Table 2. These three factors, together with the interest rate and housing stress indicator, form a 5-variable TFAVAR. I define the list of endogenous variables y_t as the interest rate as well as the Housing Stress Indicator, and the list of factors $f_t = [F1, F2, F3]$ as the first three principal components extracted.

	F1	**F2**	**F3**
% of Variance	63.936%	15.589%	13.990%
Cumulative %	63.936%	79.524%	93.515%

Table 2: Variance Explained by Principal Components

Factor loadings in Table 3 help interpret each component. F1 loads positively on real GDP, prices, real estate loans, and house prices, and negatively on mortgage rates. It captures broad real economy and housing market conditions. F2 loads heavily on the term spread and, to a lesser extent, on credit spreads, as such it can be interpreted as a yield-curve and risk sentiment factor. F3 emphasizes credit spreads and also reflects term spreads and mortgage rates, capturing broader financial stress and borrowing conditions.

	Loadings			**Contributions (%)**		
Variable	**F1**	**F2**	**F3**	**F1**	**F2**	**F3**
MR	-0.806	-0.077	0.366	14.51	0.54	13.68
CS	-0.357	0.464	0.781	2.85	19.74	62.23
TS	0.179	0.910	-0.340	0.71	75.90	11.81
HL	0.981	0.058	0.140	21.52	0.30	2.01
HP	0.906	-0.186	0.234	18.36	3.16	5.61
P	0.950	0.063	0.181	20.18	0.36	3.36
Y	0.989	-0.006	0.113	21.87	0.00	1.31

Table 3: Loadings and Contributions for Factors 1–3

With the above derived factors and the remaining variables in mind, we can write the reduced form TFAVAR as seen in Equation 2 below:

$$\begin{bmatrix} y_t \\ f_t \end{bmatrix} = \sum_{k=1}^{4} \Theta_{1,k}\, I(x_{t-1} \geq \gamma) \begin{bmatrix} y_{t-k} \\ f_{t-k} \end{bmatrix} + \sum_{k=1}^{4} \Theta_{2,k}\, I(x_{t-1} < \gamma) \begin{bmatrix} y_{t-k} \\ f_{t-k} \end{bmatrix} + u_t, \quad (2)$$

where $\Theta_{i,k}$ are the matrices of regime specific coefficients, $I(.)$ is the regime indicator, γ is the threshold value and u_t are the reduced form residuals. x_t is the threshold variable which in our case is the housing stress indicator. This specification defines a two-regime TVAR, where we can interpret one of the regimes as boom and bust periods of the housing market cycle respectively. The TVAR model is linear in parameters for a fixed threshold value γ and fixed threshold variable x_t and can be estimated using Conditional Least Squares. The threshold value is set to $\gamma = \gamma^*$, where γ^* is the optimal threshold value obtained using a grid search that minimizes the residual sum of squares.

Shocks are identified using the state-of-the-art sign restriction methodology of Arias et al. (2018). After estimating the reduced-form TFAVAR and obtaining the Cholesky factor Σ of the residual covariance matrix, I construct an orthonormal rotation matrix Q in the five-dimensional factor space. I then impose the following zero-horizon sign restrictions on the observed variables: $[R : +,\ TS : -,\ CS : +,\ MR : +,\ HL : -,\ CGG : -,\ P : -,\ HP : -,\ RP : +,\ Y : \text{no restriction}]$.[1] The structural impact matrix ΣQ is mapped back to the space of observed variables using the factor loadings, allowing for verification that the imposed signs hold. As permitted by the TVAR framework, this identification procedure is implemented on a regime-specific basis, allowing the initial Cholesky factor Σ to vary across regimes. In practice, the IRFs at the zero horizon however turn out to be near identical for most of the observed variables.

Generalized Impulse Response Analysis

A challenge in nonlinear VAR modeling is a good representation of the impulse responses. While the regime-wise orthogonalized impulse responses are relatively informative in showcasing the different dynamics across the two regimes, they do not account for the regime switches themselves. Following the work of Koop et al. (1996), GIRFs are an appropriate tool to fully account for the dynamics of TVAR models. To construct the set of GIRFs, I closely follow algorithm outlined for the Interacted-VAR model of Andreasen et al. (2021). Specifically, the GIRFs for y_t at horizon h to a monetary policy shock ϵ_{monpol} can be defined as

[1] Note that a standard FAVAR with recursive ordering already resolves the price puzzle and produces plausible responses for GDP and inflation. Here, additional sign restrictions are imposed to ensure that the contemporaneous effects on financial variables align with theoretical expectations.

$$GIRF_y(h, \epsilon_{monpol}, x_{t-1}) \equiv E_t(y_{t+h}|\epsilon_{monpol}, x_{t-1}) - E_t(y_{t+h}|x_{t-1}) \tag{3}$$

With the above definition in mind, the GIRFs depend on the regime which is defined using the relation of x_{t-1} to the threshold value γ. I construct a set of GIRFs for both regimes using the following algorithm:

- I randomly sample $K = 100$ realizations of the residuals u_t from regime 1.
- For each random draw K, I simulate the h period ahead forecast $y_{t+h}|x_{t-1}$ for $h = 0 : 20$.
- For each random draw K, I simulate the h period ahead forecast $y_{t+h}|\epsilon_{monpol}, x_{t-1}$ for $h = 0 : 20$ by adding ϵ_{monpol} to the randomly sampled realization of residual u_t, then I calculate $GIRF_y$ using Equation 3.
- I repeat steps 1-3 for $H = 100$ randomly sampled histories (starting points) from regime 1 and take the average of the obtained GIRFs $\overline{GIRF}_y(h, \epsilon_{monpol}, x_{t-1}) = \sum_1^H \sum_1^K \frac{1}{H}\frac{1}{K} GIRF_y(h, \epsilon_{monpol}, x_{t-1})$.
- I repeat the process for regime 2.

As this procedure is executed in the dimensions of the five-equation TFAVAR, I first compute the GIRFs for the latent factors, then use the estimated factor loadings (weights) to recover the responses of the original series. Using this same approach, I disaggregate the housing boom-bust cycle indicator into its two components (RP and CGG) yielding ten GIRF sets per regime. These are presented in the Results section. All variables except the policy rate (R) are shown in standardized units, reflecting the data normalization prior to factor extraction. The policy rate, which remains in levels, is displayed in percentage points. Confidence bands are constructed from 2000 bootstrap replications using the Maximum Entropy Bootstrap procedure of Vinod (2006) in order to handle non-stationary series.

Results

Is the monetary transmission truly state-dependent? A growing body of empirical work shows that both the conduct and effects of monetary policy vary with macro-financial conditions. The generalized impulse response functions (GIRFs) in

Figure 2 show that (i) the Fed holds interest rates higher for longer during periods of housing booms, and (ii) credit aggregates, real activity, and prices fall somewhat more sharply in those periods. Still, for most variables, regime differences are smaller than one might expect, especially considering the relatively wide confidence bands. Strikingly, house-price responses are nearly identical across regimes, suggesting that there is no major change in the asset price channel across the two regimes.

Nevertheless, as the financial accelerator mechanism of Bernanke et al. (1999) suggests, adverse financial market conditions amplify shocks to macroeconomic aggregates. Output, prices and credit volume appear to contract more in a booming real estate market following the monetary policy shock. Similarly to the findings of Fry-Mckibbin and Zheng (2016), the larger contraction in credit volumes is not accompanied by widening credit spreads, which respond almost identically in both regimes, suggesting no significant change in the pricing of risk across the two regimes.

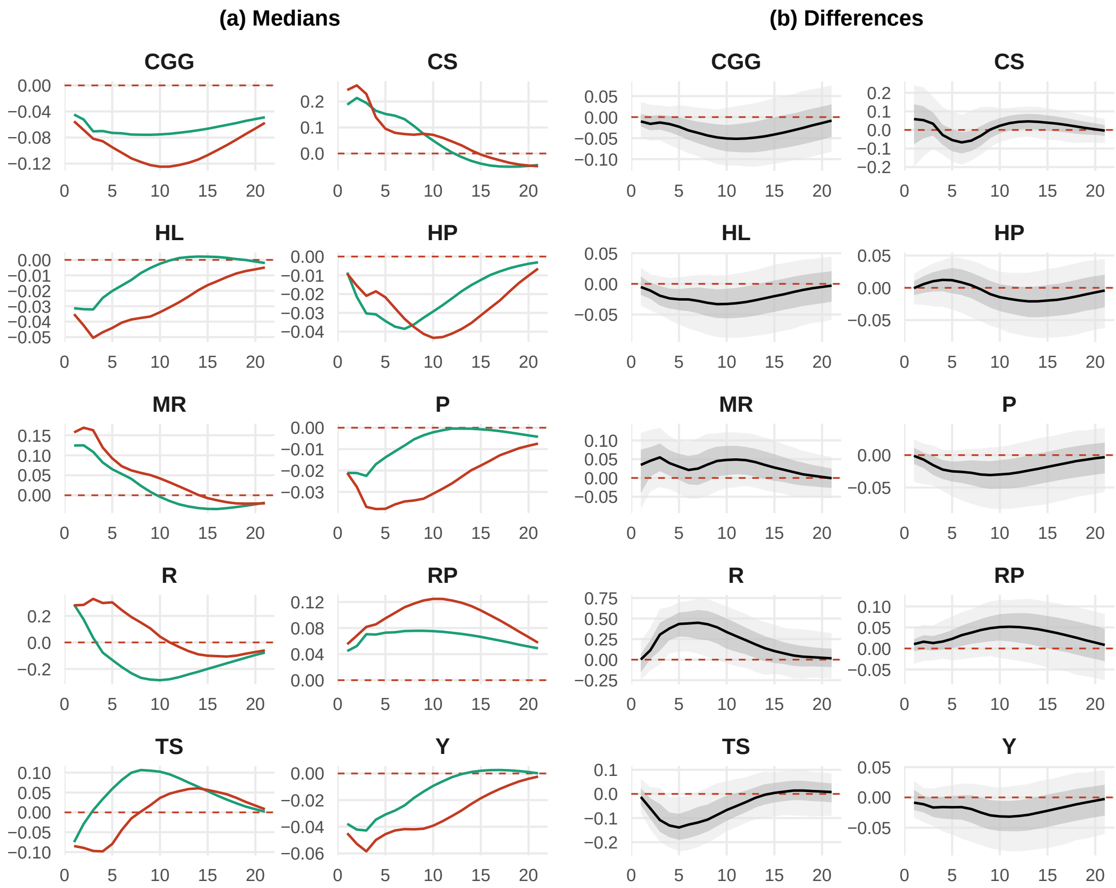

Figure 2: Generalized Impulse Responses.

Notes: Panel (a) shows the median impulse repsonses estimated from the bootstrap exercise. The color green indicates busts, while the color red indicates booms in the housing market. Panel (b) shows the differences between the regime specific impulse responses along with the 68% and 95% confidence bands in darker and lighter grey respectively.

The most substantial regime difference lies in the policy rate path (and by extension, the term spread) following a monetary tightening. In the boom regime, the Fed maintains elevated rates for several quarters before easing. By contrast, during times of housing busts, the Fed begins to reverse course more quickly. These results support Fry-Mckibbin and Zheng (2016), who argue that monetary easing is delayed when financial conditions are fragile. Similarly, Adrian and Liang (2018) argue that a looser stance during boom phases can fuel excessive leverage and risk-taking, prompting central banks to maintain tighter policy during potential housing bubbles. More recent work of Kiley and Mishkin (2024) suggests that this cautious rate adjustment reflects an explicit weighting of financial-stability objectives in the Fed's framework.

Why use the composite index instead of either RP or CGG? As discussed in the Identifying the boom-bust cycle in the Housing Market subsection, in order to successfully identify emerging bubbles in the housing market, correctly identifying high leverage along real estate market misalignments is key. Figure 3 shows the regime classifications using purely RP, purely CGG and lastly the composite index as threshold variables holding all else constant. Using purely RP as the threshold variable incorrectly identifies essentially two periods. Firstly it fails to identify the late 80s / early 90s Savings and Loans (S&L) crisis, as the excessive credit buildup preceding the real estate market adjustment was not taken into account. Secondly, it classifies essentially all of the post 2000s time periods as a period of booming real estate market, even though home prices returned close to their fundamental value as well as excessive credit buildups have substantially moderated following the GFC up until early 2020.

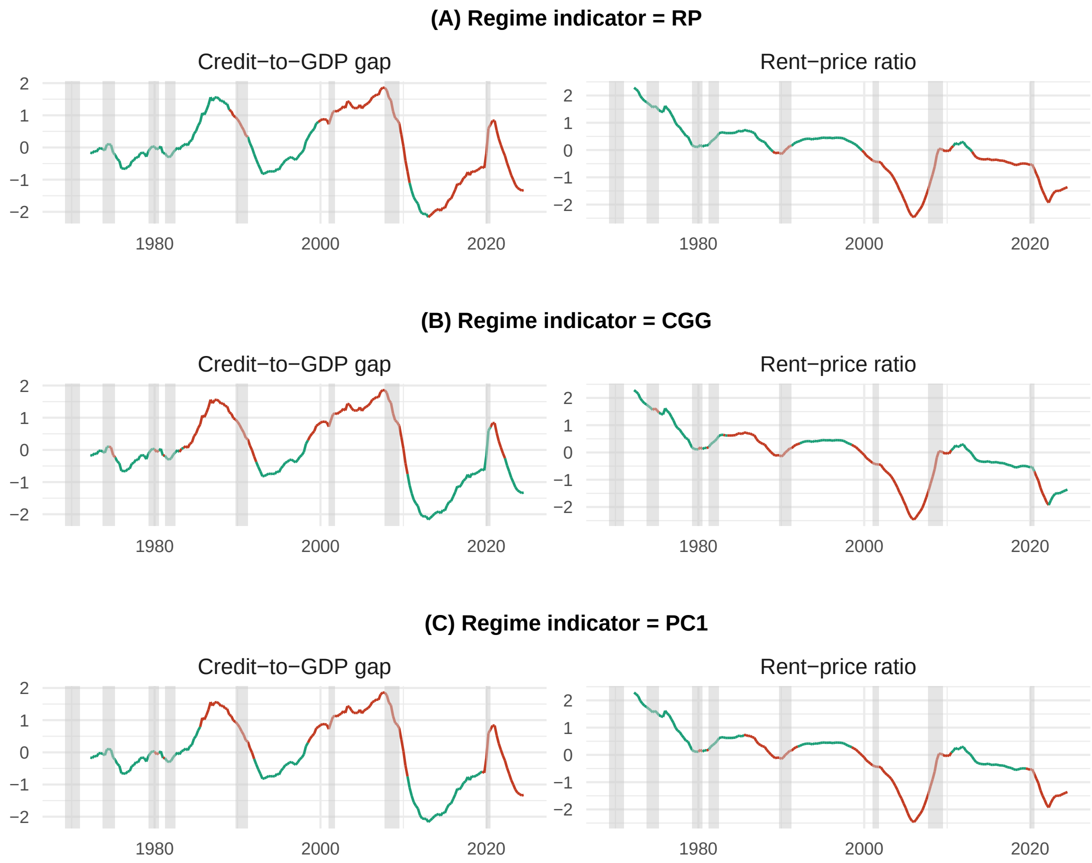

Figure 3: An indicator of housing market stress

Notes: The color green indicates busts, while the color red indicates booms in the housing market using the RP, CGG and their composite index (PC1) as regime indicators respectively. Shaded areas indicate US recessions using the Hamilton GDP-based recession indicator.

On the other hand, using purely the CGG as the threshold variable fails to take into account housing fundamentals. Around the S&L crisis, it gives "too early of a warning" along the credit cycle, way before home prices started becoming misaligned from their fundamental value. Moreover, it fails to identify housing market misalignments around the 2020s. Leveraging the linear combination of both aggregates as the threshold variable yields us regime classifications that seem to align with historical events when real estate market misalignments were likely fueled by excessive credits and overleveraging.

The model's definition of a forming housing bubble is also particularly elegant. Housing boom begins when rising credit gaps coincide with a sharp decline in rent-price ratios, and it persists until both metrics revert to their underlying trends or fundamentals. In other words, a bubble is identified whenever excessive leverage coincides with (and likely fuels) a misalignment of real estate prices from their fundamental values, ending only once the markets correct and cool off. Moreover, the regimes do not change abruptly, distinctly isolating phases of the housing cycle and yielding a continuous, timely indicator of boom-bust periods in the housing market.

Conclusion

This paper examines how U.S. monetary policy shocks propagate conditional on the state of the housing market. I construct a composite housing-stress indicator using principal component analysis on the rent–price ratio and the credit-to-GDP gap and embed it as the threshold variable in a two-regime threshold factor-augmented VAR (TFAVAR). This framework effectively identifies boom-bust cycles in the housing market. To trace the propagation of shocks, I estimate generalized impulse response functions (GIRFs) and regime-specific shocks. A bootstrap simulation exercise confirms that monetary policy itself is indeed state-dependent. What stands out most from the results is the Fed's conduct: during housing booms, the stance of monetary policy is adjusted much more carefully after an intervention. This caution reflects the central bank taking a broader set of macro-finantial variables into account when adjusting policy rates in order to avoid the possibility of a rapid resurgence in lending amid falling interest rates.

From a policy perspective, several key implications emerge. First, the response of real estate prices to monetary shocks does not differ substantially across regimes. This implies that the asset-price channel remains stable and monetary policy shocks alone are unlikely to cause a major crash. Second, while the financial accelerator appears to operate throughout the housing cycle, its amplification effects are somewhat muted. Still, this provides a rationale for pursuing active policy interventions during booms in order to restrain excessive credit growth. Third, the slower reversal of interest rates in the boom regime implies that policymakers proceed gradually when easing to avoid a resurgence of lending booms. In this sense, the results align with recent literature advocating for macro-financial considerations in monetary policymaking.

Looking ahead, several extensions could deepen this analysis. One important direction is to explore cross-sectional heterogeneity, as real estate dynamics may vary significantly across regions. This could be pursued either by extending the analysis to multiple countries or by examining state-level data within the U.S. Both approaches would pair well with a richer FAVAR framework that incorporates a broader set of variables in the factor extraction process. Finally, the composite housing-cycle indicator could be further refined by incorporating additional indicators such as loan-to-value or debt-to-income ratios. While such data are less widely available, the current index already provides a useful and interpretable signal of emerging housing bubbles.

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